

Zeros of Linear Functions from Equations

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A **linear function** is a function that can be written in the form $f(x) = mx + b$, where m and b are constants.

Zero of a Function

The **zero** of a function is the value of x that makes the function equal to zero.

Finding the Zero

To determine the **zero**, set $f(x) = 0$ and solve for x .

Zero and x-Intercept

The zero of a linear function corresponds to the **x-intercept** of its graph.

Practice Exercises

Practice 1–5

Determine the zero of each linear function.

$$f(x) = 3x - 12$$

$$f(x) = 5x + 20$$

$$f(x) = 7 - 2x$$

$$f(x) = \frac{x}{4} - 6$$

$$f(x) = \frac{3x}{5} + 9$$

Practice 6–10

Determine the zero of each linear function.

$$f(x) = 8x - 3$$

$$f(x) = 15 - 5x$$

$$f(x) = \frac{x - 8}{2}$$

$$f(x) = \frac{2x + 7}{3}$$

$$f(x) = \frac{5x - 15}{4}$$

Activities

Activity 1–5

Determine the zero of each linear function.

$$f(x) = 4x - 20$$

$$f(x) = 6x + 18$$

$$f(x) = 11 - 3x$$

$$f(x) = \frac{x}{5} - 8$$

$$f(x) = \frac{4x}{7} + 12$$

Activity 6–10

Determine the zero of each linear function.

$$f(x) = 9x - 4$$

$$f(x) = 18 - 6x$$

$$f(x) = \frac{x - 10}{3}$$

$$f(x) = \frac{3x + 5}{4}$$

$$f(x) = \frac{7x - 21}{5}$$

Thank you!